



FLBC-IDS: a federated learning and blockchain-based intrusion detection system for secure IoT environments

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Abstract

The development of the Internet of Things (IoT) in multiple sectors has led to substantial security issues due to its decentralized structure and resource limits. To solve these problems, this paper proposes a novel technique called FLBC-IDS, which combines Horizontal Federated Learning (HFL), Hyperledger Blockchain, and EfficientNet to detect intrusions in IoT environments. The FLBC-IDS model uses HFL to enable secure and privacy-preserving model training across numerous IoT devices, resulting in decentralized data privacy and resource efficiency. The model's integration of Hyperledger Blockchain ensures tamper-resistant and transparent recording of model updates and agreements among participating IoT nodes, protecting the system's integrity. Furthermore, EfficientNet improves the model's robustness by effectively obtaining and categorizing features from network traffic data. On the CICIDS-2018 and CICIOT-2023 datasets, the FLBC-IDS model outperforms previous techniques in terms of accuracy, recall, F1-score, and precision. The FLBC-IDS model has an accuracy of 98.89%, recall of 98.044%, F1-score of 98.29%, and precision of 98.44%. The article analyzes the suggested model's merits and limitations, as well as future research directions for real-world deployment of IoT security solutions. Overall, the FLBC-IDS paradigm holds significant promise for improving IoT security and safeguarding IoT settings from infiltration attempts and cyber threats.

Keywords IoT · Federated learning · Intrusion detection · Blockchain · EfficientNet

1 Introduction

The growing number of IoT devices and the trend of storing and analyzing technological interactions have contributed to a dramatic increase in data generation and storage in the digital era. This data has become an asset for companies that seek to enhance their products and services through big data and data analysis. However, with this increased data

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consumption, cybersecurity has become essential to prevent data leaks, malicious intrusions, and service interruptions [1].

According to a report by Statista, the number of monthly global IoT intrusions from 2020 to 2022 exceeded 10.54 million in December 2022. However, in December 2021, there were approximately six million fewer reported IoT attacks. The month with the highest number of attacks was June 2022, with approximately 13 million recorded [2]. From 2020 to 2022, Fig. 1 depicts the number of global IoT attacks.

Intrusion Detection Systems (IDS) are essential for monitoring system activity and proactively detecting potential attacks. First-generation IDS relied on signature databases but lacked flexibility and proactivity, rendering them incapable of rapidly detecting new threats. Consequently, developing novel Machine Learning (ML) models has been closely tied to the evolution of threat detection techniques [3]. Support Vector Machine (SVM) and Random Forests (RF) are fundamental ML models incorporated into second-generation intrusion detection systems to learn attack patterns and automatically enhance detection accuracy. Simultaneously, Deep Learning (DL) methodologies such as Multilayer Perceptron and Recurrent Neural Networks increased complexity. To promptly perceive and respond to new attack vectors, parties must share threat information in real-time. Algorithms for collaborative learning with strong data protection regulations provide a scalable infrastructure that respects privacy for efficient threat detection [4].

Federated Learning (FL) has become a promising method for managing information interchange between parties and addressing privacy concerns regarding sensitive data. Due to its low communication costs and privacy-preserving capabilities, this advanced ML technique has acquired considerable interest in edge and IoT computing [5]. Although FL's current application to IDS technologies is limited, it will be the foundation of future IDS advancements. Google introduced FL in 2016 [6], FL promotes collaborative learning by enabling local detection with low latency while protecting anonymity. In contrast to Collaborative IDS (CIDS) techniques, FL mitigates local bias in ML-based IDS and alleviates concerns about latency and bandwidth [7]. Combining FL with blockchain technology can further improve data privacy, low-latency network communication, and learning

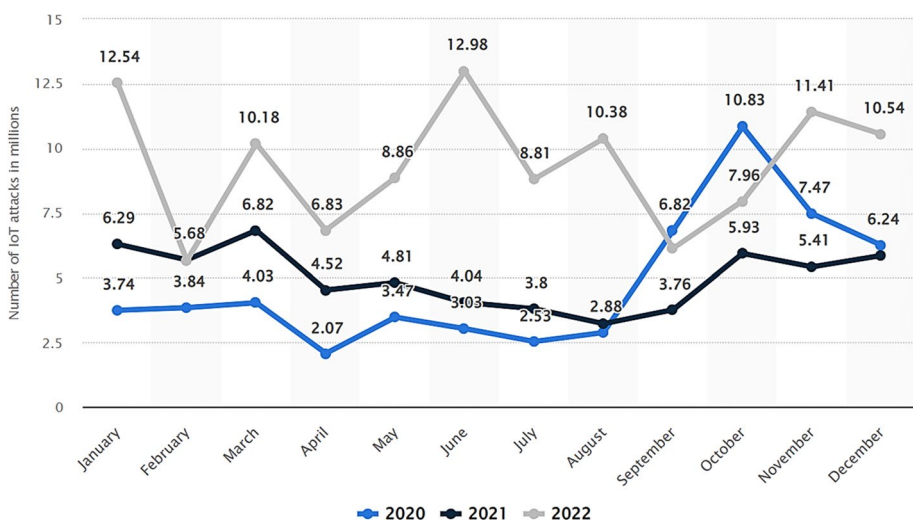


Fig. 1 Global IoT Attacks from 2020 to 2022

quality, thereby enhancing the cyber defences of IoT devices and networks. Three kinds of FL include:

- Horizontal FL: Data sets share the same feature space, but the sampling space is distinct.
- Vertical FL: Data sets with different feature spaces but the same sampling space.
- Federated transfer learning: Data sets have varying characteristics and sampling spaces [8].

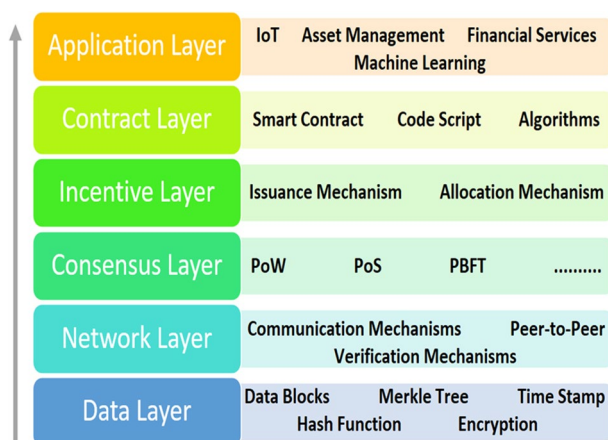
The Federated Learning Systems (FLS) encompass various elements of FL taxonomy, including learning models, data partitioning, privacy, and communication features. FLS can handle numerous ML models, ranging from basic linear to complex neural networks. K-Anonymity modifies datasets to guarantee anonymity, and Differential privacy adds noise to secure individual records. FL possesses two paradigms: Global server for centralized model aggregation and asymmetric communication. Decentralized—Model parameters are shared via point-to-point communication (P2P, graph, blockchain), posing trust and convergence challenges [9].

Agent characteristics determine federation size: Cross-Silo—Few parties with substantial computational resources, such as data centers and corporations. Cross-device—Multiple parties with limited computational capabilities, such as IoT devices and smartphones. The data partitioning methods with their IID and non-IID variants are as follows: Horizontal- Distinct parties with minimal sample overlap (non-IID). Vertical- Agents with comparable samples but distinct characteristics. Both feature and sample spaces vary between agents in hybrid models [10]. Motives for FLS include Incentives—Parties profit from precise models without sharing valuable information. Compliance with data restrictions and laws promotes model sharing for collaborative research (e.g., hospital medical data) [11].

IDS is a widely employed security technology, and while some IDS implementations utilize DL for enhanced precision. However, insufficient data hinders model training in some institutions. The centralization of DL raises privacy concerns. FL helps surmount these obstacles by enhancing IDS accuracy through collaborative model training while securing data privacy [12].

Blockchain is a novel technology that uses cryptography to save information in a distributed ledger. It enables multiple entities to share and verify information transparently. In the blockchain, smart contracts implement predefined conditions. Some studies examine the use of blockchain and smart contracts for IDSs. Combining FL and blockchain guarantees the safe sharing of local model updates between IoT endpoints. This combination paves the way for blockchain and FL-based IoT applications.

Figure 2 depicts the six layers of the blockchain architecture. The data layer includes transaction records from various sources in Merkle tree blocks with timestamps, hashes, and verified transactions. Each block in the chain has a preamble containing metadata and a hash of the preceding block, forming the chain. Blockchain's network layer facilitates P2P decentralized interactions between nodes. It conducts transmission, verification of transactions, and broadcasting. Nodes sign transactions with their private keys and broadcast them to their neighbours, who validate them using their public keys. While valid transactions are incorporated into blocks and broadcast to other nodes, invalid transactions are discarded. Blockchain's consensus layer ensures ledger efficiency, security, and integrity among untrusted peer-to-peer network nodes. Its primary objective is to reach a consensus on integrating a newer block into the ledger. Several consensus models are utilized in the blockchain framework, each with advantages and disadvantages. The incentive layer provides

Fig. 2 Blockchain Architecture

economic incentives in blockchain systems, particularly public blockchains. It issues and distributes digital currencies to nodes that verify new blocks, fostering decentralization. The contract layer contains script codes, algorithms, and smart contracts that augment transactions with sophisticated business rules. When predefined conditions agreed upon by nodes are met, these rules execute automatically. The blockchain application layer contains a variety of applications that utilize its features. Examples include AI, IoT, security systems, financial systems, and cloud computing. Bitcoin's application layer consists of digital currency transactions, whereas Ethereum and Hyperledger Fabric platforms include currency transactions and decentralized applications [13].

This study uses Federated Learning (FL) to provide private and secure model training across several IoT devices while maintaining data confidentiality and efficiency without exchanging raw data. FL's distributed nature improves the model's stability and adaptability, which are critical for effective intrusion detection in IoT systems. Furthermore, blockchain technology, specifically Hyperledger, is used to protect model revisions and agreements, resulting in a tamper-proof ledger for increased integrity. The suggested paradigm is critical in the face of rising cyber threats to IoT ecosystems, as it addresses standard intrusion detection systems' limitations. By combining FL and Blockchain, the concept provides collaborative device training while protecting privacy and prohibiting illegal changes. This approach improves the overall security and reliability of intrusion detection in IoT contexts, which is critical for protecting sensitive data and reducing possible hazards.

FL's effectiveness for IoT is dependent on factors such as algorithm complexity, model complication, and device resources. Feasibility can vary and be classified as:

- **High-End IoT Devices (Level 1):** These IoT devices have strong computing power (for example, industrial gateways or advanced sensors) and can perform moderately complex FL tasks effectively.
- **Mid-Range IoT Devices (Level 2):** These devices, such as standard cameras and smart thermostats, have moderate computational capacity. For simplified models and lightweight algorithms, FL works well.
- **Low-End IoT Devices (Level 3):** These resource-constrained IoT devices (e.g., simple sensors) have limited power and memory, rendering FL challenging. Typically, only fundamental model updates and data preprocessing are possible.

In the context of computationally intensive intrusion detection using the datasets and the EfficientNet model, it is generally more feasible to implement FL on IoT devices classified as Level 1 or Level 2 devices. These devices are more likely to be capable of running the required DL model and effectively collaborating in the federated training process. Due to their limited computational resources, Level 3 IoT devices may not be suited for executing DL models such as EfficientNet.

1.1 Problem statement & scope of the research

This research's problem statement is to develop and implement an IDS for IoT that addresses data privacy, security, and collaboration challenges. When applied to sensitive IoT environments, the centralized nature of traditional IDSs may raise concerns regarding data exposure and privacy violations. The research proposes using HFL and a Hyperledger Blockchain to address these issues. This study seeks to develop an HFL and Hyperledger Blockchain-based IDS for IoT employing the EfficientNet DL model on the CSE-CIC-IDS2018 dataset. The system will employ HFL to permit IoT devices or networks to train an EfficientNet DL model for intrusion classification. A Hyperledger Blockchain will exchange model updates securely and transparently, ensuring data privacy and immutability.

The novelty of the research includes FLBC-IDS, a novel intrusion detection system that integrates HFL, Hyperledger Blockchain, and EfficientNet to provide robust security for IoT environments. HFL ensures decentralized data privacy and resource efficiency, whereas Hyperledger Blockchain ensures an immutable ledger. EfficientNet improves the extraction and classification of features. Evaluation using both the CICIDS-2018 and CICIOT2023 datasets is significant. This dual-dataset evaluation methodology ensures that the model is evaluated against a variety of intrusion scenarios, demonstrating its adaptability and efficiency.

1.2 Research objectives

A novel FL and blockchain-based IDS (FLBC-IDS) is proposed to handle IoT network data security and privacy. The FLBC-IDS model is proposed to break the isolated architectures by allowing learning over distributed big data. The objectives of this research are as follows:

- To implement the HFL and Hyperledger blockchain-based IDS model for IoT.
- To apply the EfficientNet model for training and testing and classifying intrusions.
- To enhance data privacy and security and network communication with low latency.
- To speed up the training and reduce communication when compared to centralized learning.
- To evaluate the proposed model with real-time and benchmark datasets, which are used for attack detection and classification regarding data privacy and security.
- To validate this proposed model with existing models based on accuracy, detection rate, precision, and f1-score.

The article is ordered as follows: Section 2 discusses reviewing the related works based on FL and Blockchain-based IDS. Section 3 explains the implementation of the proposed model, including HFL, Hyperledger blockchain and the EfficientNet model. Section 4

evaluates the performances of the research model by comparing it with existing models, and Section 5 discusses the conclusion and future works.

2 Related works

2.1 Review of FL-based IDS

To classify the intrusions, an FL approach for IDS in Industrial IoT networks was proposed in [14] using both RNN and CNN models. FL ensures privacy by training local IoT data locally and sharing parameter updates with a central server. This work can concentrate on bolstering dependability against malicious nodes, detecting outliers, and selecting devices with adequate resources. Device failures and sluggish model updates are obstacles that influence the global model's accuracy. In [15], a Segmented-FL for adaptive IDS in large-scale LANs was created. Network segmentation and periodic local model assessment were utilized to group comparable environments. Using deep CNN and expert knowledge-based labelling, it outperforms conventional FL in intrusion detection tasks and improves detection performance by sharing parameters among group members.

A multi-view ensemble FL IDS for IoT was proposed in [16]. It coupled multi-view learning with FL to efficiently detect and classify attacks. This model utilized FL to ensure decentralized training without sharing raw data to identify, classify, and secure attacks. The grey wolf optimizer was used to optimize the extraction of features. Using a random forest classifier, the ensemble-based technique incorporated predictions from three perspectives for greater accuracy. In [17], a federated DL scheme for cyber threat detection in industrial cyber-physical systems was devised. The approach combined a DL-based IDS employing CNN and GRU with an FL framework for privacy-preserving collaboration between multiple cyber-physical systems. To protect the confidentiality and safety of model parameters during training, a safe protocol of communication for employing the Paillier cryptosystem was implemented. The results of the experiments demonstrate the efficacy and superiority of this model. This model was customized for industrial CPSs within the same domain to develop a federated IDS model.

The research in [18] emphasizes improving the performance of IoT anomaly detection for FL. The researchers address issues associated with data distribution across devices and class imbalance in FL. To enhance detection performance, they investigate data augmentation techniques employing five distinct FL systems: random oversampling, stratified oversampling, SMOTE, ADASYN, and GANs. Stratified and uniform random sampling provides the greatest performance lift with the least computation time. In contrast, GANs are the most time-consuming and provide the least benefit. The study in [19] examined and assessed FL for intrusion detection in the IoT. In the past, centralized approaches have been utilized for ML-enabled IDS in IoT, raising privacy concerns. FL has emerged as a privacy-preserving alternative, but its implementation in IoT IDS is still in its infancy and requires additional study. This research evaluates an FL-enabled IDS model using a multiclass classification and distributions from the ToN_IoT dataset. The impact of various aggregation functions is assessed utilizing the IBMFL model as the FL deployment.

FedIoT, a cybersecurity infrastructure for FL in IoT, was proposed in [20]. Deep Autoencoder was utilized for anomaly detection, and FedDetect, an FL algorithm with adaptive optimization and cross-round learning rate, was proposed. FedIoT supports IoT periphery and CPU/GPU-based distributed training, efficiently detecting various attack types on IoT

devices with limited resources. Using a probabilistic hybrid ensemble model, the research in [21] developed an FL-based IDS model for IoT security. Using weighted convex surrogate loss functions, the work also proposed a noise-tolerant hybrid ensemble model in federated and centralized contexts to address the label-noise problem. The results on noisy data demonstrate that the noise-tolerant hybrid ensemble model effectively manages symmetric label noise while maximizing the true positive rate and minimizing false positives. The investigation in [22] examined FL for malware detection in IoT devices. It contrasts supervised and unsupervised federated models with conventional approaches and presents a framework using the N-BaIoT dataset. Federated and centralized methods with diverse and extensive data are more efficient and protect participant privacy. An adversarial configuration reveals a weakness in the baseline model aggregation, whereas other aggregation functions are more resistant to malicious attacks. Integrating blockchain technology into the framework can improve client security and resolve privacy concerns.

FL assures privacy for user data, adding a safe layer to IoT networks and improving their dependability. The work in [23] proposed a decentralized FL model with an ensemble model for anomaly identification in IoT networks. It allows device-based training and reduces data transfers to a centralized server. LSTM and GRU neural networks achieve efficient ML model training on the Modbus dataset. Compared to centralized ML, the results demonstrate enhanced intrusion prediction and reduced false alarms. FL-Aided CNN was proposed in [24] as an intelligent IDS for edge-assisted IoT using FL. With the attention mechanism, the model ensured data privacy, avoided transferring raw information to a central server, and achieved a 99.76% classification accuracy for attack data. The work in [25] used FL to detect intrusions in the Industrial IoT while protecting privacy. This work evaluates differential privacy techniques on non-independent data and demonstrates that Fed+ maintains accuracy even in noise, ensuring acceptable accuracy with minimal privacy requirements. FL-based Attention GRU (FedAGRU) was proposed in [26] as an enhanced FL algorithm for intrusion detection in wireless edge networks. FedAGRU attained greater precision, robustness, and efficiency than Federated Averaging (FedAVG) and is highly resistant to poisoning attacks.

In [27], a new data poisoning attack for FL-based IoT IDSs was described, revealing vulnerability to covert attacks by injecting weaknesses into detection models, resulting in misclassification of harmful traffic. The attack goes undetected and bypasses FL poisoning safeguards in legitimate IoT datasets. Meanwhile, [28] created a federated deep reinforcement learning-based anomaly detection system for IIoT that achieves accurate detection at high throughput and low latency while maintaining user privacy.

2.2 Review of FL and blockchain-based IDS

In [29], suggested a Blockchain and FL-based cloud IDS paradigm for distributed detection of malicious assaults in IoT networks, leveraging FL and blockchain for secure data sharing and model training information storage. A module for identifying alert filters was added to improve the FL model's accuracy while also protecting privacy, reducing client burden, and delivering appropriate feedback. However, issues include optimizing RSP (Regional Service Party) alarm analysis in real-world circumstances and developing a white/blacklist approach to effectively control intrusion emergencies and reduce bandwidth strain caused by false alarms.

Meanwhile, [30] worked on creating an autoencoder model for anomaly detection in FL, recreating normal patterns to detect anomalies in dispersed data, and assuring

accountability and auditing capabilities while maintaining data privacy. Although FL and blockchain address privacy and security problems, issues like free-rider and model poisoning attacks need time-consuming model auditing, which raises communication costs. To address this, [31] proposed a security-based solution that uses a double-layer aggregation model with competing vote verification methods and aggregation algorithms to reduce communication costs while maintaining model security and IoT node availability.

Furthermore, [32] proposed a FL model based on hierarchical blockchain for collaborative IoT-IDS, which ensures secure and confidential detection by sharing cyber threat intelligence across IoT networks. The hierarchical FL architecture, along with a cloud-fog-edge topology, improves security and allows for quick identification of aberrant activity while protecting data privacy. Furthermore, [33] and [34] demonstrated blockchain-based FL systems for IoT devices that ensure privacy, secrecy, traceability, and precision in detecting threats to key IoT infrastructures. These research jointly highlight the potential of FL and blockchain technology to enhance IoT security while addressing various difficulties and limits.

2.3 Review summary

Existing works on FL and blockchain-based IoT intrusion detection propose various approaches that combine FL and blockchain technologies to accomplish secure and privacy-preserving IoT network intrusion detection. These approaches use distributed data training to protect user privacy while achieving high detection accuracy for malicious activities. Integrating blockchain technology guarantees accountability and traceability by providing a secure, immutable ledger for recording model updates and transactions. Existing works utilized DL models such as CNN, GRU, LSTM, and autoencoders for anomaly detection. Several works address issues such as outlier detection, reliability against malicious nodes, and device failure management. In addition, methods for addressing noise, label-noise issues, and poisoning assaults are evaluated and proposed. The research demonstrates that FL and blockchain can enhance the dependability and security of IoT-IDSs, with potential enhancements focusing on optimizing alarm analysis, implementing blockchain mechanisms, and further enhancing model robustness against attacks.

3 Proposed IDS model

This research proposes an IoT-based intrusion detection system using HFL, Hyperledger Blockchain, and the EfficientNet model. The IDS enables multiple IoT devices to train an EfficientNet model without sharing sensitive data or communicating with one another. HFL ensures privacy and resource efficiency. The model modifications are securely recorded on the Hyperledger Blockchain, which ensures their integrity and prevents tampering. On the CSE-CIC-IDS2018 dataset, the trained EfficientNet model is evaluated for its efficacy in detecting and mitigating intrusions. This integrated methodology provides a highly secure, privacy-preserving, and dependable IDS to protect IoT environments from cyber threats.

3.1 Data preprocessing

Data preprocessing is crucial in intrusion detection, which entails analyzing network traffic to identify malicious activities or anomalies. Data preprocessing is an essential step in the

development of ML models. It is crucial in IoT attack datasets, which can be extensive and complex. The proposed preprocessing strategies address problems in IoT attack datasets by maintaining data integrity, reducing bias, and improving model generalization. Cleansing removes irregularities and aberrations from data, allowing for more accurate analysis of malicious activity. Scaling normalizes feature scales, avoiding bias against specific features and allowing the model to effectively learn from a variety of IoT threat situations. This helps prevent bias towards certain features during training and enhances the machine learning model's performance. This procedure was carried out using the min–max scaling normalization technique. Normalization is a data preprocessing technique that scales the values of numerical features to lie within a similar range, typically between 0 and 1. The objective of the normalization procedure is to ensure that each feature contributes the same quantity to the machine learning model, regardless of the initial measurement scale. Using the following formula, the min–max scaling algorithm transforms the value of each feature, x , into a new value, x_{norm} .

$$x_{norm} = \frac{(x - \min(x))}{\max(x) - \min(x)} \quad (1)$$

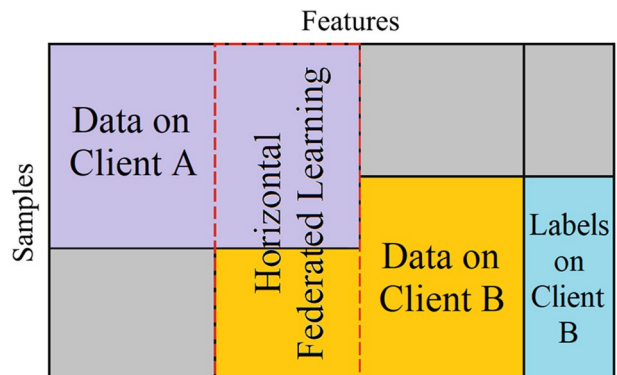
In this context, $\max(x)$ and $\min(x)$ represent the minimum and maximum values of the feature x in the data set, respectively. x_{norm} , the final normalized value will lie between 0 and 1 [36]. This procedure separates the dataset into training and test sets. In a data set representing an IoT attack, it is crucial to ensure that the model does not improperly match the data and can generalize well to previously unseen data. In this instance, the datasets were divided such that 80% would be used for training the model and 20% for testing it.

3.2 Horizontal federated learning

HFL is designed for clients with datasets containing identical features but distinct samples. The term "horizontal" is derived from instance-distributed learning, in which the entire data set is horizontally structured across data instances and allotted to two clients. As depicted in Fig. 3, data can also be horizontally partitioned in FL when various clients (devices) generate data with the same attributes (features).

There are three distinguishing characteristics between instance-distributed learning and HFL. Data is typically IID in distributed learning, whereas in HFL, it may be non-IID. While distributed learning can manually assign client data subsets to be IID for speedier

Fig. 3 Structure of HFL



concurrence, HFL's central server lacks access to raw client data, which may not be IID. Second, HFL requires numerous connected clients, whereas distributed learning typically relies on fewer clients to maintain distributed training performance. Lastly, the mechanisms for updating global models differ marginally. In instance-distributed learning, like multi-GPU training, a deep neural network (EfficientNet) modifies the global model synchronously following calculating mini-batch data local gradients. Due to communication cost constraints, this approach is less suitable for HFL, as frequent data upload and download are undesirable [37]. Figure 4 depicts the proposed HFL model's key steps.

Step 1: The central server initialized and distributed the global model parameters to all connected devices. This assures that each device begins with the same initial model state, fostering consistency throughout the training process.

Step 2: All linked devices train the global model independently on their unique local datasets for numerous iterations. Each device computes the updated model's gradients or parameters using local data and processing resources. These updates are then relayed to the central server, where they are aggregated in Step 3.

Step 3: for efficiently updating the global model, the central server collects updates from all devices, either synchronously or asynchronously, and generates an updated version of the global model. This aggregation procedure ensures that the model accurately reflects the pooled knowledge gained from all participating devices.

Step 4: Repeat steps 2 and 3 until convergence is attained.

Throughout this procedure, some characteristics are critical for ensuring efficient communication and privacy. The global model parameter (θ) reflects the current state of the model, while N denotes the total number of devices. n_k Indicates the size of data

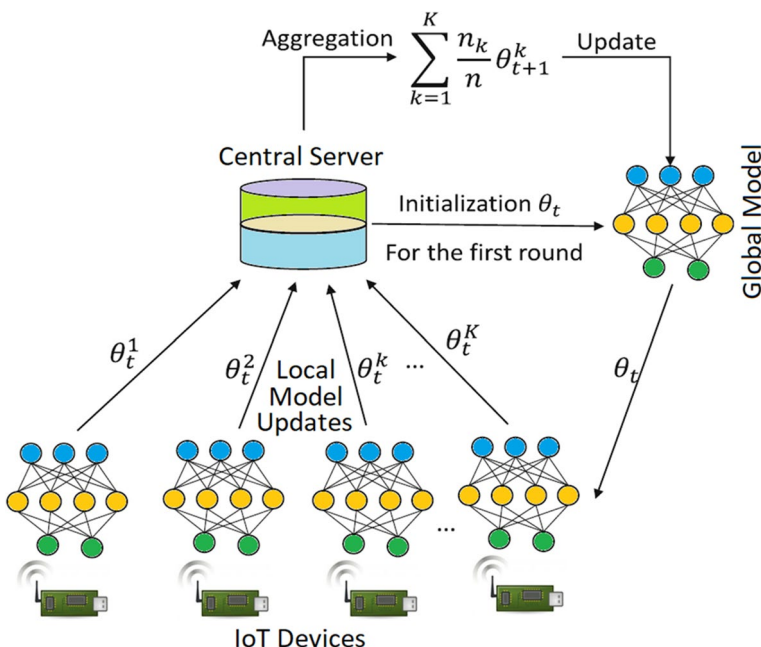


Fig. 4 Flowchart of the Proposed HFL Model

on device i . additionally, the communication round represents each iteration of communication between the devices and central server. HFL's privacy protection system is an important factor to note. HFL consumers' privacy is strongly protected by guaranteeing that the central server only receives model weights or gradients from linked devices, not raw data. This method reduces the risk of revealing sensitive data during the model training phase, hence increasing the trust and security of the HFL framework. This ensures that the privacy of HFL consumers is strongly protected. HFL protects data privacy using a decentralized architecture that keeps sensitive information localized on individual devices. By sending model updates or gradients rather than raw data, HFL prohibits the central server from accessing sensitive information, protecting HFL consumers' privacy. Additionally, encryption techniques are used during data transfer to improve security. This privacy-centric approach ensures that users' data is kept secure throughout the collaborative model training process."

HFL enables collaborative training of the EfficientNet model across several IoT devices while maintaining data privacy. Its decentralized design assures that raw data is secure, alleviating privacy worries. HFL outperforms other federated learning approaches in terms of resilience and adaptability since it allows devices with varied datasets to collaborate in model training. This results in a more complete and representative model for intrusion detection in IoT contexts.

3.3 EfficientNet

The EfficientNet model extracts high-level features from network traffic data that has been preprocessed. The EfficientNet model will aid in identifying significant patterns and abstract representations within the raw data. The final classification was performed using a CNN architecture based on the selected features [38]. EfficientNet is a model scaling technique designed to accelerate CNNs by scaling each dimension uniformly using a compound coefficient. It accomplishes a balance among network characteristics to enhance performance overall. EfficientNet, which was created with AutoML and uses MBConv as its primary structural component and swish activation function, outperforms other intrusion detection models due to its efficient scaling approach, which optimizes performance while not significantly increasing computational resources. Compared to topologies such as MobileNetV2 and MnasNet, EfficientNet strikes a better balance between network properties, resulting in improved precision and resource utilization. Its capacity to effectively extract features from raw data makes it an appealing option for intrusion detection in IoT contexts, providing increased accuracy and efficiency. The general architecture of EfficientNet consists of sixteen MBConv, one average pooling, two convolutions, and a classification layer. The MBConv module is essential, as it implements the attention mechanism of the SE network to enhance training time, performance, and random depth. It enables effective feature extraction by concentrating on essential data and suppressing irrelevant features.

EfficientNet is a highly effective model that makes efficient use of computational resources. As a result of compound scaling, the model group now consists of eight models, B0 through B7, with enhanced precision and without a significant increase in parameter count. As described in Eq. (2), the composite coefficient adjusts depth, width, and resolution uniformly to estimate the coefficients for composite proportions.

$$\begin{cases} \text{depth} : d = \alpha^\phi \\ \text{width} : w = \beta^\phi \quad \text{s.t.} \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2 \\ \text{resolution} : r = \gamma^\phi \end{cases} \quad (2)$$

Together with the variables w , d , r , the coefficients $\alpha \geq 1$, $\beta \geq 1$, $\gamma \geq 1$ are used to calculate the efficient resource extension model and constants α , β , γ for network width, depth, and resolution.

The research suggests that the optimal network coefficients are $\gamma = 1.15$, $\beta = 1.1$, $\alpha = 1.2$. Due to the convolution operation's unique computational overhead, the convolution network's scalability based on Eq. (2) will enhance the network's total FLOPS.

The MBConv is the fundamental building block of the EfficientNet framework, derived from the MobileNet model concepts and modified to satisfy MBConv's specifications. It includes depth-separable point and depth-separable convolution layers that are applied sequentially. The MobileNet-v2 inverted residual connection and linear bottleneck are combined in MBConv. Convolution refers to the weighted superpositions of functions, denoted by the notation $r(x)$, which represents the overlap length integral of overlapping functions, f and g , formed by shifting and rotating.

$$r(x) = \int_{-\infty}^{\infty} f(\tau)g(x - \tau)d\tau \quad (3)$$

Through convolutions, the system's output was obtained, with function $f(x)$ representing actual pixel points and function $g(x)$ representing action points. Each action point was utilized to perform additional processing on convolutional kernels. The final convolution result was obtained after all action points were implemented on the convolutional kernels and combined with actual pixel points. Through an element-wise operation on the input vector denoted ' T ', the ReLU function converted nonnegative values from the convolutional layer to zero [43].

$$\sigma(I) = \max(0, I) \quad (4)$$

The CNN's pooling layer (downsampling layer) minimizes parameters and features by halving the matrix volume. Typically located at the top of a CNN, this layer summarizes the characteristics of the layers below it. The photos are then input into a feed-forward neural network, facilitating the discovery of nonlinear functions. Standardized data are incorporated into the training procedure to distinguish between important and lower-level image features. Methods such as Softmax are utilized for classification. The dropout layer is represented by Eq. (5).

$$Z_i^L = w_i^L y^L + b_i^L \quad (5)$$

In the above equation, L represents a hidden layer, Z_i^L the center layer, y^L the vector output, w_i^L the weights, and b_i^L the biases in this instance.

Softmax assigns a cost to neurons, with a maximal load of one for a single neuron and zero for all others. The softmax function is denoted by Eq. (6).

$$f(x_i) = \frac{e^{x_i}}{\sum_{j=1}^N e^{x_j}} (i = 1, 2, \dots, N) \quad (6)$$

Using the Softmax function, the final layer of the neural network determines the likelihood that each input belongs to a particular category. The EfficientNet model contains manually added layers such as two convolution layers, batch normalization, dropout,

and pooling layers [39]. The model employs an SGD optimizer with a 0.001 learning rate to attain optimal performance.

For intrusion detection, the datasets are formatted and converted using a combination of LSTM and EfficientNet:

Initial tokenization of dataset data, such as log entries and network traffic descriptions, into individual words or subword units. Using a specified padding token, these tokenized sequences are then padded or truncated to a fixed length, typically 100 tokens per sequence, to ensure consistency. To facilitate DL, tokens are converted to numeric representations utilizing pre-trained word embeddings such as Word2Vec. The data is then organized into a 3D tensor of the shape (batch_size, sequence_length, embedding_dimension), which is an input format suitable for DL models. Initially, converted image data from the dataset is imported. For uniformity, these images are resized to a consistent size, typically 224×224 pixels, and their pixel values are normalized to lie within the interval [0, 1]. The images are then grouped together for further processing or analysis. Simultaneously, the EfficientNet branch processes the image data, preprocessed image tensors, including essential preprocessing steps. A classification layer is then added on top of these features, typically consisting of one or more densely connected layers. In the concluding step, a softmax activation function is applied to facilitate classification [44].

3.4 Hyperledger Blockchain

In HFL, the central server is typically viewed as trustworthy and inquisitive, adhering to the FL protocol while endeavouring to infer data information. The gradients shared by each device have the potential to cause data leakage. Consequently, this study employs Hyperledger Blockchain as a decentralized and tamper-resistant ledger to document and store model updates and agreements among the participating IoT devices. After each iteration of local training on an IoT device, a model update is generated and propagated to a specific channel in the Hyperledger Blockchain network. The consensus mechanism is used to agree on the update's validity, assuring that all participating nodes execute a consensus algorithm to verify its validity. Once consensus is attained, the model update is combined with others to create a new block in the blockchain, which provides an immutable and transparent record of the model training process. This decentralized approach improves the IDS's security, integrity, and accountability, making it suitable for reliable and privacy-preserving intrusion detection in IoT environments [45].

Permissioned blockchain technology is used by hyper-ledger fabric, where the organization grants network access. It uses digital certificates to establish the identity of each node. Innovative modules like multi-ledger, multi-channel, consensus mechanisms, and rights management offer unique approaches to blockchain-based services. Private data isolation by distinct organizations and channels increases data security. Endorse Peers, Commit Peers, and Orderer Peers are peer entities in the Fabric network. Based on the consensus mechanism, Endorse Peers simulates transactions, which Orderer Peers then sort and distribute for confirmation in the global state database. A node employs a private key for digital signature when initiating a transaction in Hyperledger Fabric, employing the Elliptic Curve Digital Signature Algorithm (ECDSA), which is more efficient than RSA while maintaining the same level of security [40]. This assures the security of Hyperledger Fabric blockchain network transactions.

ECDSA: Assume the private and public keys are denoted by k and K . K is calculated as $K = kG$, where a point on the elliptic curve E is represented as G . Before the plaintext M is encrypted, a random number r was generated.

Private Key Signature:

$\emptyset = rG(x, y); h = \text{Hash}(M);$

$$s = \frac{(h + kx)}{r} \quad (7)$$

$\text{Commit Peer} \leftarrow \{M, (\emptyset, s)\}$

Peer Public Key Verification Signature:

Compute $h = \text{Hash}(M);$

$$\psi = \frac{hG + xK}{s} \quad (8)$$

Verify $\emptyset = \psi$

Introducing a random number ' r ' during the signature generation procedure in ECDSA is essential for enhancing the signature's security. This ' r ' is combined with the private key to generate a transaction-specific signature. Indeed, the resulting signatures will be unique if the same message is signed multiple times with various random numbers ' r '. This function adds a security layer, making it more difficult for potential adversaries to forge or manipulate signatures, enhancing the system's overall security. This property of ECDSA ensures that the signature remains unique and unpredictable for every transaction, preventing impersonation by unauthorized parties. Scalability concerns and network latency are two potential drawbacks of incorporating blockchain into the IDS concept. Ensuring regulatory compliance and overcoming interoperability issues among different blockchain networks are also critical factors for successful adoption.

4 Experimental analysis

4.1 Dataset

The dataset was chosen because it accurately represents modern network incursions and includes a diverse range of real-world attack techniques. With its extensive feature set and broad use in cybersecurity research, it serves as a solid foundation for training predictive models in intrusion detection, ensuring comparability and benchmarking for evaluating the performance of our proposed system. The CSE-CIC-IDS2018 dataset was developed specifically for training predictive models to detect network traffic intrusions. It is a component of an extant initiative that generates modern, applicable datasets in a scalable manner. This evaluation dataset is accessible to the public and is utilized for testing intrusion detection systems. The dataset contains data on both benign and malicious network traffic. These malicious attacks include Brute Force Password Guessing, Distributed Denial of Service, and Web Application Attacks. Using a simulated network environment, real-world traffic was collected, and the dataset was constructed. This data set consists of raw packet capture data and preprocessed features that faithfully represent the network traffic encountered in real-world scenarios. The preprocessed features include the flow duration, the number of packets and bytes, and a variety of statistics derived from the packet payloads. This data set contains 89 traffic characteristics,

ranging from the most fundamental to the most advanced, such as packet lengths, protocol types, flow, timestamps, and content-related characteristics. This dataset, which can be downloaded from the CIC website, has been utilized extensively by intrusion detection and cybersecurity researchers.

The CICIOT2023 dataset contains real-world IoT attacks that simulate installations in smart households, as well as 105 genuine IoT devices with common setups. With 33 unique assaults divided into seven categories, including Distributed Denial of Service (DDoS), Denial of Service (DoS), Reconnaissance (Recon), Web-based attacks, Brute Force attacks, Spoofing attacks, and Mirai attacks, it provides a wide range of scenarios for examination [41]. This dataset adds to the study by providing a realistic setting for testing the suggested IDS model's efficacy in identifying various IoT-related threats, increasing its relevance and usefulness in solving modern cybersecurity concerns. The dataset's distribution of network traffic is depicted in Table 1 [35].

4.2 Experimental setup

This section presents a detailed analysis of the experimental results of the proposed FLBC-IDS model. The proposed model was implemented and tested using the PYTHON 3.7.12 programming language tool. The experiments are performed on Google Colab Pro, with a computer with a 64-bit CPU, an i7 processor, and 12 GB of RAM. Various metrics such as accuracy, recall, precision, and F1-score are used to evaluate the research model's performance. The metrics are computed based on true positives, true negatives, false positives, and false negatives. These computed performances are then compared with existing models for validation.

This investigation examines the concept of client homogeneity within the context of HFL for intrusion detection in the IoT. Consider, in the context of FLBC-IDS, a network of IoT devices deployed in a smart city environment, with the objective of the proposed model being the detection of various types of intrusions and cyber threats [42]. For Example: Client 1 represents a subset of IoT devices responsible for smart city traffic management. Client 2 represents an additional subset of IoT devices deployed across the smart city in healthcare facilities.

Table 1 Dataset Distribution

Dataset				
CICIOT2023 Traffic Distribution		CICIDS-2018 Traffic Distribution		
Traffic Types	Distribution	Traffic Types	Training	Testing
Benign	83.070%	Benign	138,542	39,837
Botnet	1.763%	Recon-PortScan	11,379	1952
Brute Force	2.347%	Recon-OSScan	14,965	980
DoS	4.031%	Recon-HostDiscovery	18,448	3286
DDoS	7.786%	Recon-PingSweep	335	2
Infiltration	0.997%	Mirai-greeth_flood	130,796	29,478
Web Attack	0.006%	Mirai-udpllain	115,633	28,787
		Mirai-greip_flood	95,402	27,056

Client 1 has features related to traffic camera feeds, vehicle sensor data, and traffic flow patterns, for instance. Client 2 was assigned capabilities pertaining to patient vital signs, medical record access notes, and data transmission within healthcare facilities.

In this FLBC-IDS research, ten clients representing a variety of IoT devices deployed in a smart city participate in the federated learning procedure [43]. This clientele consists of sensors for traffic management, healthcare monitoring, environmental data collection, and surveillance.

To maintain a balanced federated learning process, a consistent selection ratio of around 30% is applied to each round. This means that roughly 30% of available clients are selected at random to participate in model updates and contribute their insights during each federated learning round.

Learning Rates: The following learning rates are used to optimize model convergence and training efficiency:

- Initial client update learning rate is set to 0.1.
- A decay factor for the learning rate of 0.90 is used to govern the learning rate decline over time.
- To prevent excessive reduction, a minimum learning rate of 0.001 is imposed.
- The global model aggregation learning rate is consistently maintained at 0.05.

The FLBC-IDS model employs the Federated Averaging with Secure Aggregation (FASA) algorithm for aggregation. FASA ensures that model updates from participating clients are aggregated in a secure manner that protects their privacy. This method employs sophisticated cryptographic techniques, including secure multi-party computation (MPC) and homomorphic encryption, to maintain the confidentiality and integrity of the federated model aggregation method.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

$$Recall = \frac{TP}{TP + FN} \quad (10)$$

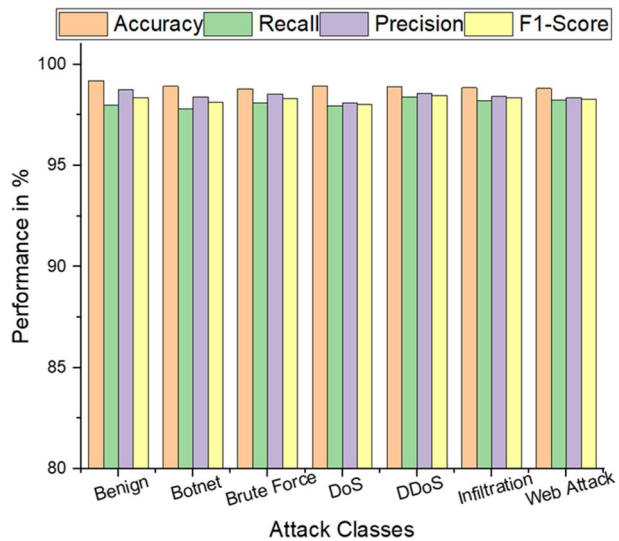
$$Precision = \frac{TP}{TP + FP} \quad (11)$$

$$F1score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (12)$$

Using the CICIDS-2018 dataset, the proposed FLBC-IDS model demonstrated strong intrusion detection efficacy for multiple attack classes in IoT environments. The performance evaluation results for the proposed model are shown in Table 2. The accuracy is the extent to which the model's predictions are accurate. As shown in Fig. 5, the FLBC-IDS model obtained high levels of accuracy ranging from 98.80% to 99.20% for various attack classes. This demonstrates that the model accurately classifies network traffic into attack categories. Recall, also referred to as sensitivity or true positive rate, quantifies the model's ability to correctly identify positive instances (intrusions) out of actual positive instances. The model obtained recall values between 97.80% and 98.40%, indicating that it can detect most intrusion attempts. Precision measures a model's ability to identify true positive

Table 2 Proposed Model's Performance Evaluation Scores on CICIDS-2018 Dataset

Attack Classes	Accuracy	Recall	Precision	F1-Score
Benign	99.20	98.00	98.75	98.37
Botnet	98.92	97.80	98.40	98.15
Brute Force	98.80	98.10	98.55	98.30
DoS	98.95	97.95	98.10	98.02
DDoS	98.90	98.40	98.58	98.47
Infiltration	98.85	98.22	98.42	98.35
Web Attack	98.81	98.25	98.35	98.29

Fig. 5 Graphical Plot of Proposed Model's Performance on CICIDS-2018 Dataset

instances among all reliably predicted positive instances. The FLBC-IDS model obtained precision values between 98.10% and 98.75%, indicating that it is extremely likely to be accurate when the model predicts an intrusion. The F1-score is the harmonic mean of precision and recall, providing a measure of a well-balanced model's performance. The F1 scores of the FLBC-IDS model ranged from 98.02 to 98.37%, indicating a reasonable balance between precision and recall.

The FLBC-IDS model demonstrated outstanding performance in classifying normal and benign traffic with high accuracy, precision, and F1-score, thereby effectively minimizing false positives. In addition, it demonstrated excellent performance in detecting botnet, brute force, DoS, DDoS, infiltration, and web attacks, attaining high accuracy and precision with low false positives. Even though botnet detection had a slightly lower recall, the FLBC-IDS model is robust and reliable in identifying different attack classes in IoT environments.

The model obtains a remarkable 98.59% accuracy on average, demonstrating its ability to correctly classify network traffic into attack and benign classes of the CICIoT-2023 dataset. The model exhibits high recall rates across all assault classes, ranging between 97.33% and 98.07%. This indicates that the model detects instances of numerous attack types effectively. The FLBC-IDS model exhibits high precision, with scores ranging between 97.74%

Table 3 Proposed Model's Performance Evaluation Scores on CICIoT-2023 Dataset

Attack Classes	Accuracy	Recall	Precision	F1-Score
Benign	98.91	97.61	98.20	97.86
Recon-PortScan	98.53	97.33	97.97	96.92
Recon-OSScan	98.42	97.72	98.12	97.85
Recon-HostDiscovery	98.64	97.54	97.70	97.66
Recon-PingSweep	98.57	98.05	98.24	98.17
Mirai-greeth_flood	98.49	97.76	98.06	98.02
Mirai-udpplain	98.52	97.82	97.92	97.59
Mirai-greip_flood	98.66	97.67	98.09	97.83

Table 4 Comparison of Performance Evaluation Scores

Models	Accuracy	Recall	Precision	F1-Score
FL-CNN [14]	93%	93%	94%	93%
FL-RNN [14]	94%	95%	95%	94%
MV-FLID [16]	94.17%	93.42%	93.26%	94.14%
FL-STRAT [18]	95.94%	92.15%	89.85%	90.99%
FL-RAND [18]	94.65%	90.24%	86.49%	88.33%
FL-GRU [23]	98.67%	96.72%	96.25%	96.48%
FedACNN [24]	98.73%	83.92%	94.67%	88.97%
Proposed Model	98.89%	98.04%	98.44%	98.29%

and 98.27%. This implies that when the model predicts an attack, it is highly likely to be accurate. The F1-Scores in this Table 3 range between approximately 97.66% and 98.16%, indicating a effective balance between precision and recall for various attack classes.

The performance of the FLBC-IDS model was compared to that of existing IDS models, including FL-CNN, FL-RNN, multi-view federated learning intrusion detection (MV-FLID), FL-Stratified Sampling (STRAT), FL-uniform random sampling (RAND), FL-Gated Recurrent Unit (GRU), and FL Aided-CNN (FedACNN), as shown in Table 4. Based on the comparison depicted in Fig. 6, the FLBC-IDS model obtained the highest accuracy of 98.89% among all models, indicating that it can classify network traffic with exceptional accuracy. FedACNN (98.73%) and FL-GRU (98.67%) also demonstrated high accuracy, closely following the proposed model, whereas other models had accuracy scores between 93% and 95.94%. The FLBC-IDS model obtained a recall of 98.04%, demonstrating its ability to accurately identify a high proportion of actual positive instances (intrusions) among all positive instances in the dataset. FL-GRU also demonstrated a high recall rate (96.72%), demonstrating its ability to detect most intrusions. Other models, such as MV-FLID, FL-CNN, and FL-RNN, obtained recall values greater than 93%, whereas FedACNN's recall was only 83.92%. The FLBC-IDS model obtained a precision of 98.44%, indicating that it is highly likely to be accurate when predicting an intrusion. FL-GRU had the second-highest precision (96.25%), just behind the precision of the proposed model. FL-CNN, FL-RNN, and MV-FLID attained precision values greater than 93%, whereas FedACNN had a lower precision of 94.67% and FL-STRAT and FL-RAND scored less than 90%. The FLBC-IDS model obtained the highest F1 score of 98.29%, a balanced

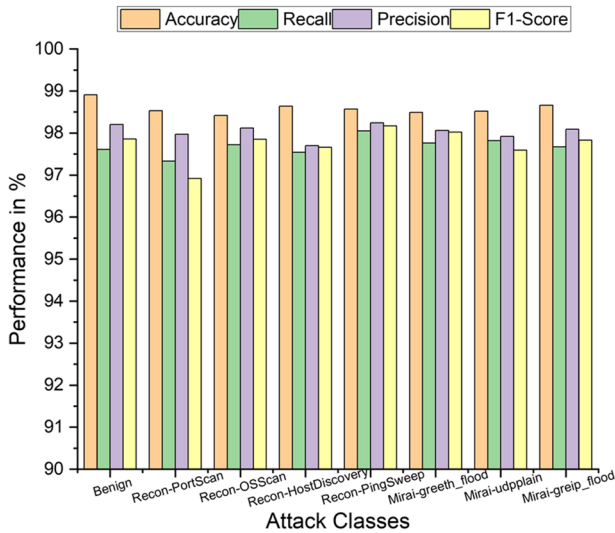
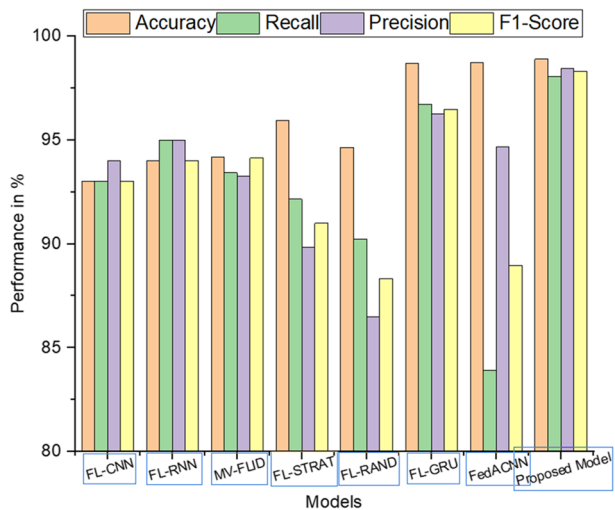


Fig. 6 Graphical Plot of Proposed Model's Performance on CICIoT-2023 Dataset

measure of accuracy and recall. FL-GRU had the second-highest F1-score (96.48%), closely trailing the performance of the proposed model. FL-CNN, FL-RNN, and MV-FLID attained F1-scores greater than 94%, whereas FedACNN's F1-score was 88.97%, and FL-STRAT and FL-RAND's F1-scores were less than 91%.

Overall, the proposed FLBC-IDS model outperforms the other extant models in accuracy, recall, precision, and F1-score is shown in Fig. 7. Its ability to achieve high accuracy while maintaining a balance between recall and precision makes it an exceptional choice for identifying and delineating attack classes in IoT environments. Incorporating HFL, Hyperledger Blockchain, and EfficientNet into the proposed model contributes to its

Fig. 7 Comparison of Performance Analysis



exceptional performance, assuring privacy, integrity, and efficient feature extraction, resulting in dependable IDS.

4.3 Advantages of the FLBC-IDS model

Privacy-preserving collaboration HFL enables IoT devices to collaborate in a secure and privacy-preserving manner. It enables devices to locally train models using their data and share only model updates instead of raw data. This decentralized strategy guarantees data privacy and confidentiality, making the FLBC-IDS model compliant with data protection regulations and boosting user confidence.

Enhanced security with blockchain Integrating Hyperledger Blockchain guarantees a tamper-proof and immutable ledger to record model modifications and agreements between IoT devices. Blockchain technology ensures the model's integrity by providing transparency and accountability, making the training process secure and resistant to malicious alterations.

Robust model with EfficientNet The EfficientNet architecture integrates EfficientNet for feature selection and CNN for classification to produce a robust model. The FLBC-IDS model is more robust and adaptable to real-world scenarios due to EfficientNet's superior performance in managing diverse and complex data.

High accuracy and reliability Compared to existing models, the FLBC-IDS model obtained high accuracy, precision, recall, and F1-score values. Its exceptional performance ensures accurate and dependable intrusion detection, minimizing false positives and negatives, essential for effective threat mitigation.

Resource efficiency Utilizing HFL, the model takes advantage of the distributed nature of IoT devices, maximizing their computing capacity while minimizing the need for centralized data storage and processing. This approach optimizes the utilization of limited resources in IoT environments.

4.4 Limitations of the FLBC-IDS model

Communication overhead HFL necessitates continuous communication between IoT devices throughout the model training process. Depending on the communication infrastructure, this may introduce communication overhead, especially in IoT environments with limited bandwidth and constrained resources.

Model complexity and training time EfficientNet models can be computationally intensive to train, particularly when working with large-scale IoT datasets. Training time may increase if IoT devices have limited computational capabilities.

Scalability of blockchain Hyperledger Blockchain, while secure, may encounter scalability challenges as the number of IoT devices participating in the network grows. As the size of the blockchain network grows, the time and resources required to reach a consensus on model modifications may increase.

Heterogeneity of data In an actual IoT environment, data collected from various devices may exhibit heterogeneity in terms of data quality, distribution, and characteristics. It is difficult to ensure model generalizability across disparate IoT datasets.

Blockchain overhead Using a blockchain to record model updates may introduce additional computational and storage overhead, which could affect the FLBC-IDS model's overall efficacy, particularly on resource-constrained devices.

5 Conclusion

In this study, we presented the FLBC-IDS model, a novel approach to intrusion detection in IoT contexts. We ensured secure and private model training across IoT devices by utilizing Horizontal Federated Learning (HFL), without jeopardizing raw data privacy. Furthermore, the incorporation of Hyperledger Blockchain enabled an auditable and transparent record of model revisions and agreements, which improved the training process's security, integrity, and accountability. The EfficientNet model, with its efficient feature extraction capabilities and compliance with sequential data shows effectiveness for network traffic classification in an ID setting. Our experiments on the CIC-IDS2018 and CICIOT-2023 datasets indicated that the FLBC-IDS model outperformed previous benchmarks. With an excellent precision of 98.89%, recall of 98.044%, F1-score of 98.29%, and precision of 98.44%, the FLBC-IDS model demonstrated exceptional performance in identifying and mitigating intrusion attempts in IoT environments. However, we must admit our study's shortcomings. While our model produced encouraging findings, there may be issues with its performance optimization, scalability, and robustness in real-world deployment scenarios. Addressing these issues will be critical to developing the area of IoT security and assuring the long-term resilience of IoT ecosystems against cyber threats. Future research should focus on improving the FLBC-IDS model's performance, scalability, and applicability to various IoT contexts. By solving these issues, we can help to create safer and more reliable IoT ecosystems, ultimately protecting vital infrastructure and maintaining the security and privacy of IoT users worldwide.

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Data availability Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

Declarations

Ethics approval and consent to participate No participation of humans takes place in this implementation process.

Human and animal rights No violation of Human and Animal Rights is involved.

Conflict of interest Conflict of Interest is not applicable in this work.

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